

Optimal Design of Water Distribution Systems

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Bridging Science to Practice



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Optimal Design - WDS

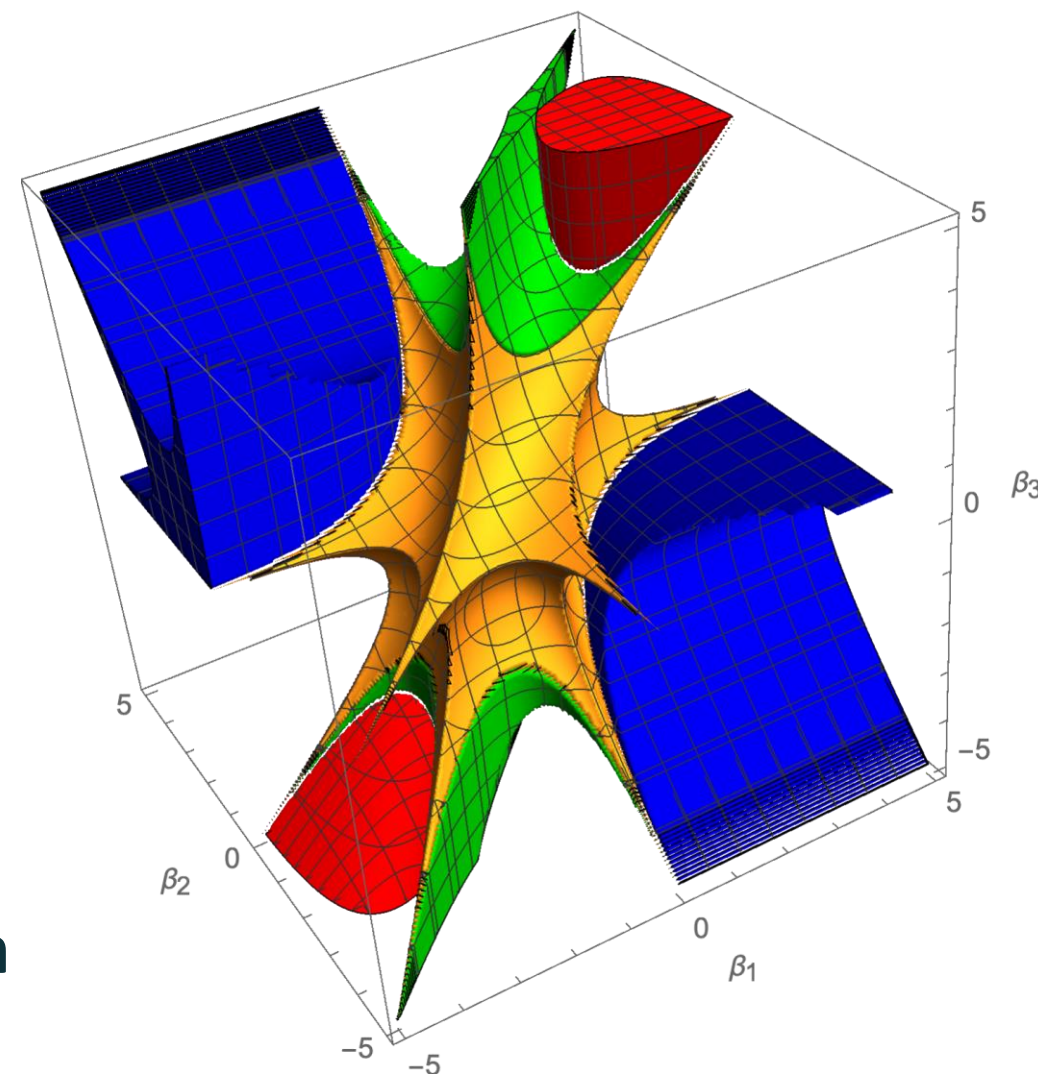
Why does optimal design matter?

- Cost efficiency, resilience, reliability, sustainability

Increasing pressures on WDS

- Ageing infrastructure, climate uncertainty, urban growth

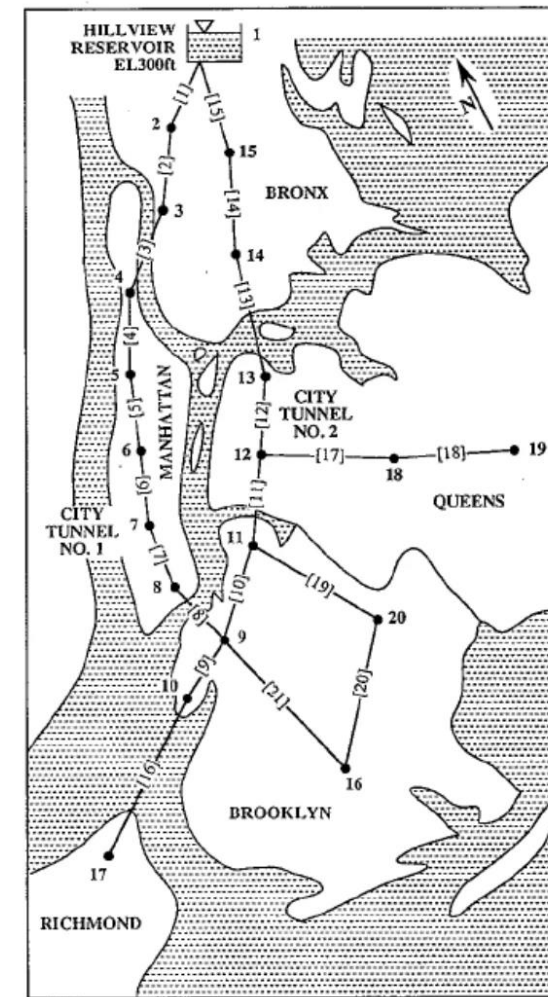
The shift from “design by rules” to “design by optimisation”



Types of WDS design problems

Most WDS are existing systems, hence most often dealing with:

1. Expansion (e.g., new area being developed)
2. Strengthening (e.g., ensuring better performance of the existing systems)
3. Rehabilitation (e.g., replacing less efficient elements with new ones)
4. A combination of the above



New York Tunnels problem

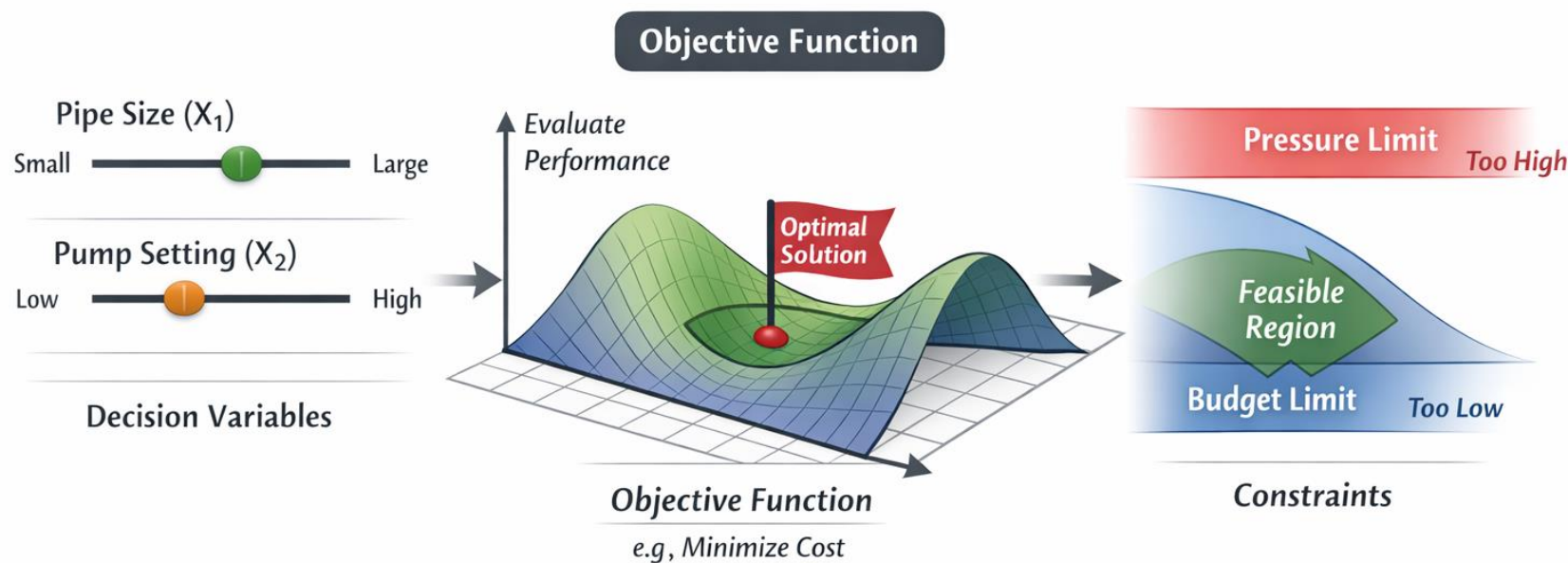
Optimisation of WDS: Basic Concepts

What is optimisation?

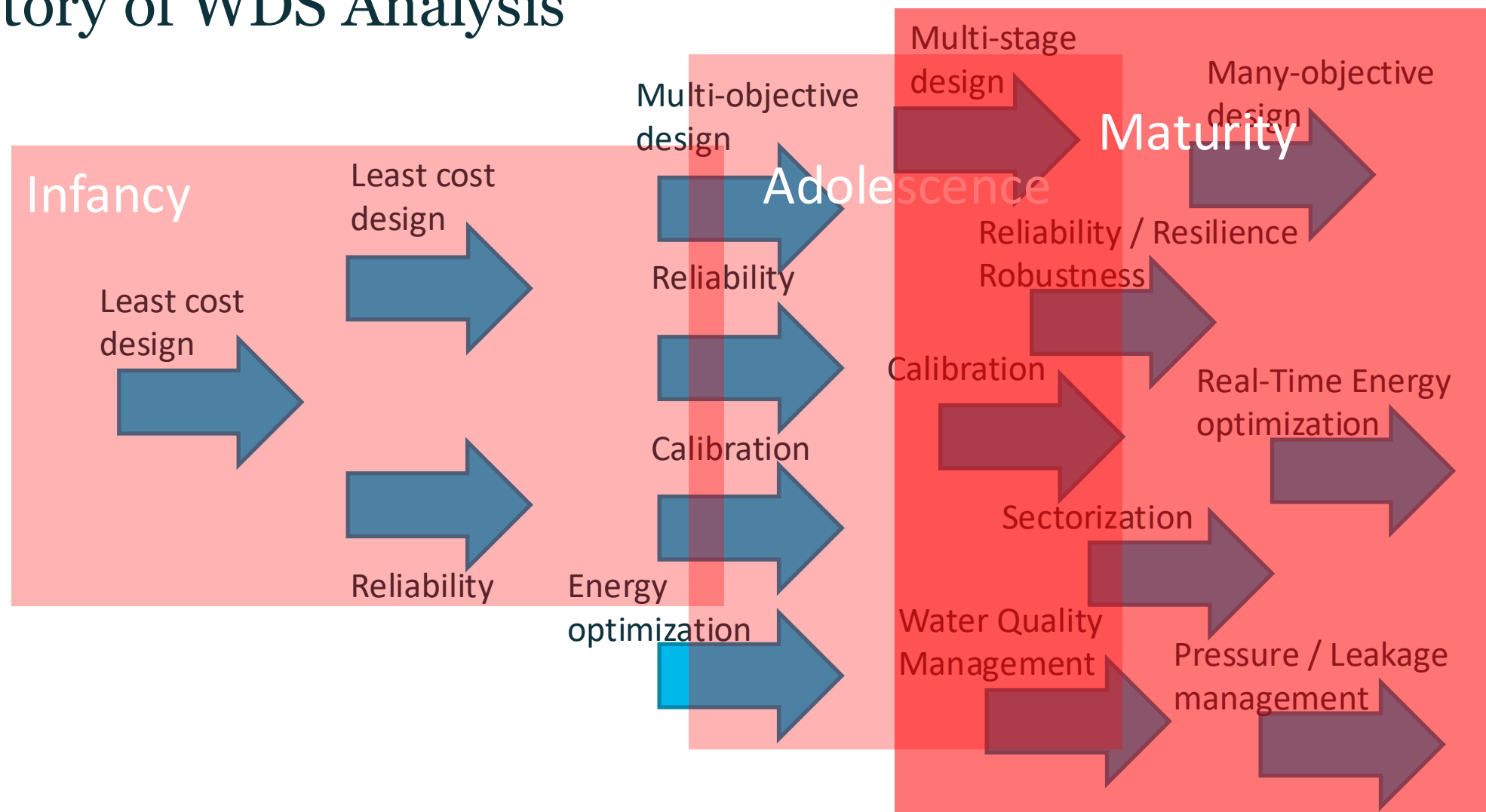
- Optimisation is a smart way of finding the best possible answer to a problem when you have many choices and can't try them all by hand

Optimisation formulation

- Objective functions
- Decision variables
- Constraints



History of WDS Analysis



Optimisation of WDS: Basic Concepts

Typical objectives

- Cost minimisation
- Pressure performance
- Energy use
- Resilience

Common decision variables

- Pipe diameters
- Pump schedules
- Valve settings
- Network layout

Feasibility and hydraulic constraints

- Pressure
- Flow
- Tank levels
- Energy balance

Simulation–optimisation loop

- EPANET or similar hydraulic solvers embedded in optimisation

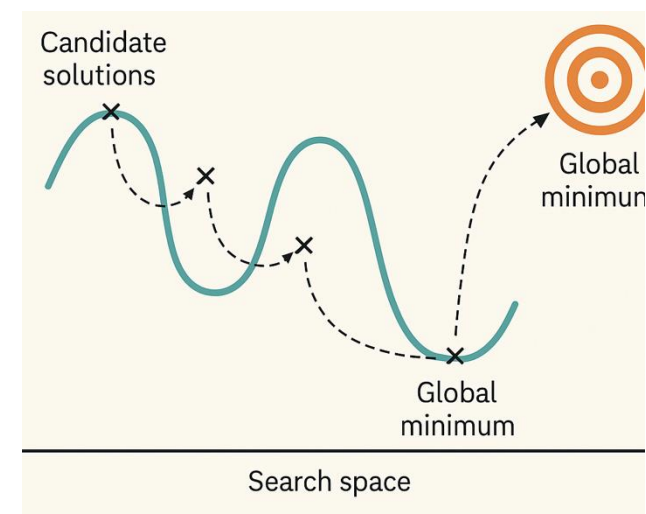
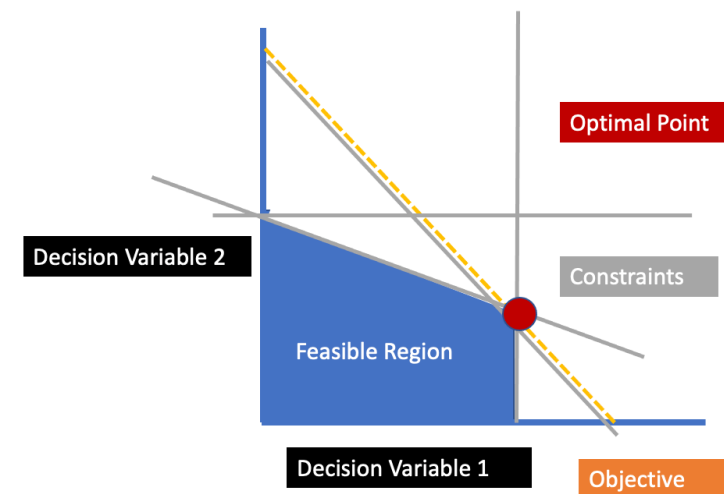
Classical Optimisation Techniques

Deterministic methods

- Linear programming, nonlinear programming
- **Strengths:** guaranteed optimality
- **Weaknesses:** nonconvexity, discrete decisions, complex hydraulics

Metaheuristics

- Genetic Algorithms (GA), Simulated Annealing (SA), Particle Swarm Optimisation (PSO), Harmony Search, Ant Colony, etc.
- **Strengths:** Flexible, handle discrete variables, global search
- **Weaknesses:** Computationally expensive, sensitive to parameter tuning



Traditional Approaches to WDS Design

Design a water distribution system that

- satisfies key performance indicators (KPI)
- provides robust performance over a number of scenarios
- is both efficient in terms of capital investment (CAPEX) and operational costs (OPEX)
- (takes into account staged development)



Traditional Approaches to WDS Design

1. Model of existing system (if any)

2. Layout of proposed piping

3. Preparation for model runs

a) demand estimates

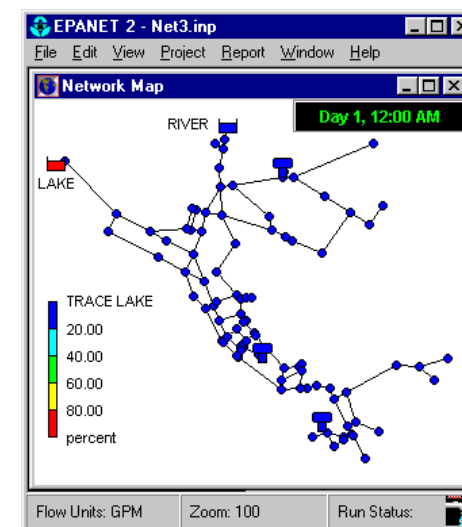
b) initial pipe size estimates $D = \sqrt{\frac{C_f Q}{V}}$, $C_f = 1274$ for Q in l/s, D in mm and V in m/s

c) possible outages to check reliability

4. Model Runs

a) Epanet

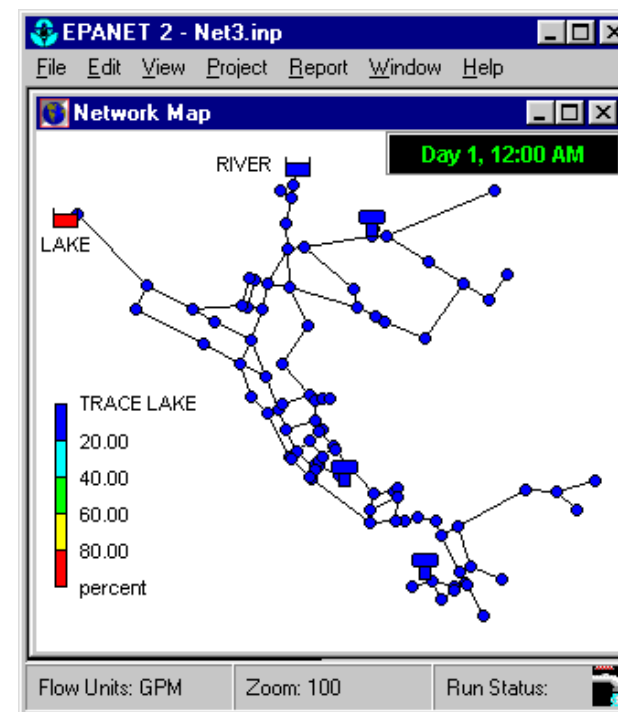
b) or any other software



Traditional Approaches to WDS Design

4. Model Runs (continued) to check for

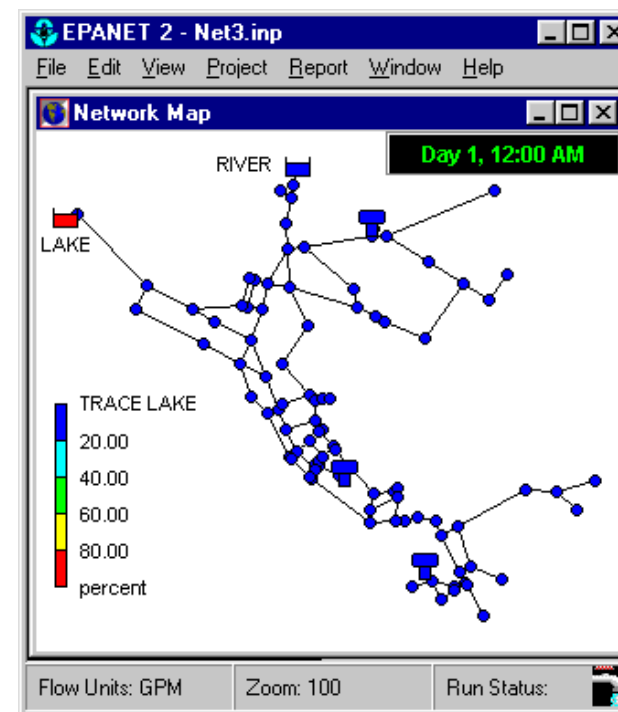
- a) High velocities
- b) Pressure below minimum
- c) Pumps not operating at desirable points on pump curve
- d) Tanks not draining or filling at desirable rates
- e) Unusually high pressures
- f) Low velocities during peak demand periods
- g) Low chlorine (disinfectant) residuals or high water age



Traditional Approaches to WDS Design

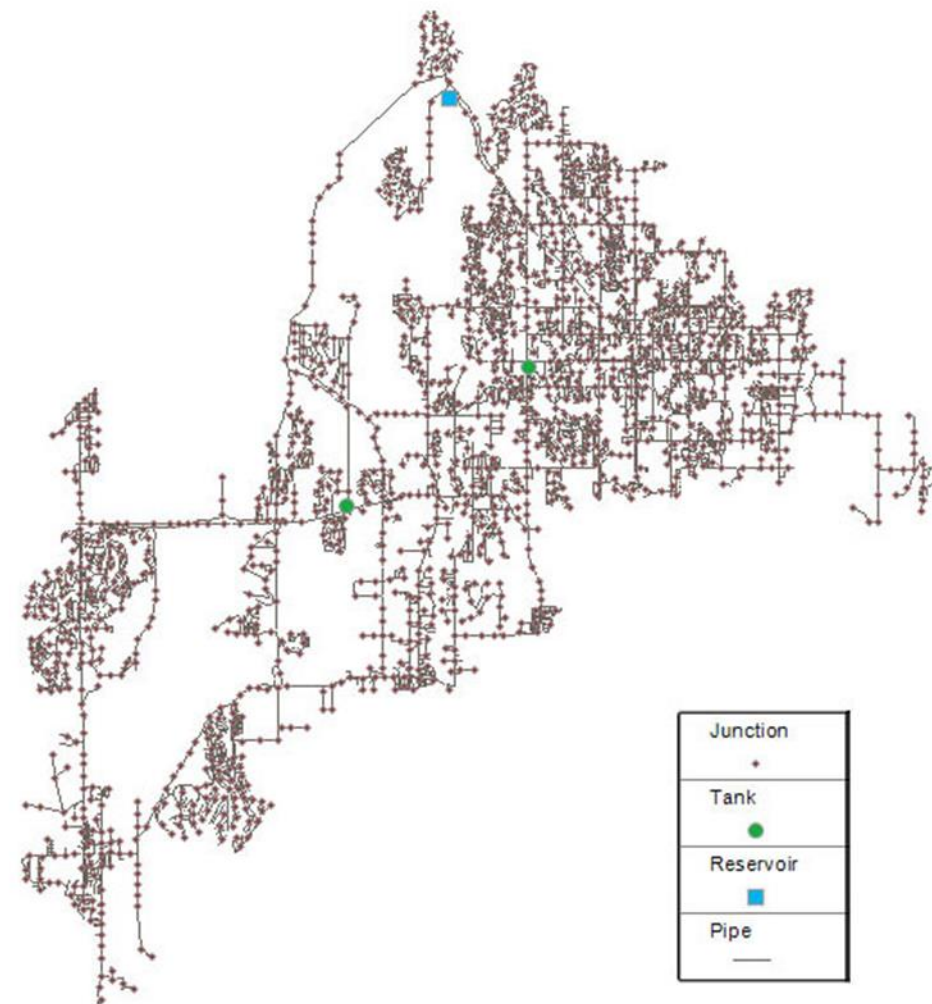
5. Compare with Standards and Guidelines

- a) if not satisfied – reformulate design and go back to step 4 (**Model Runs**), otherwise
- b) estimate costs (CAPEX and OPEX)
- c) stop if found best cost, otherwise reformulate design and go back to step 4 (**Model Runs**)



Traditional Approaches to WDS Design

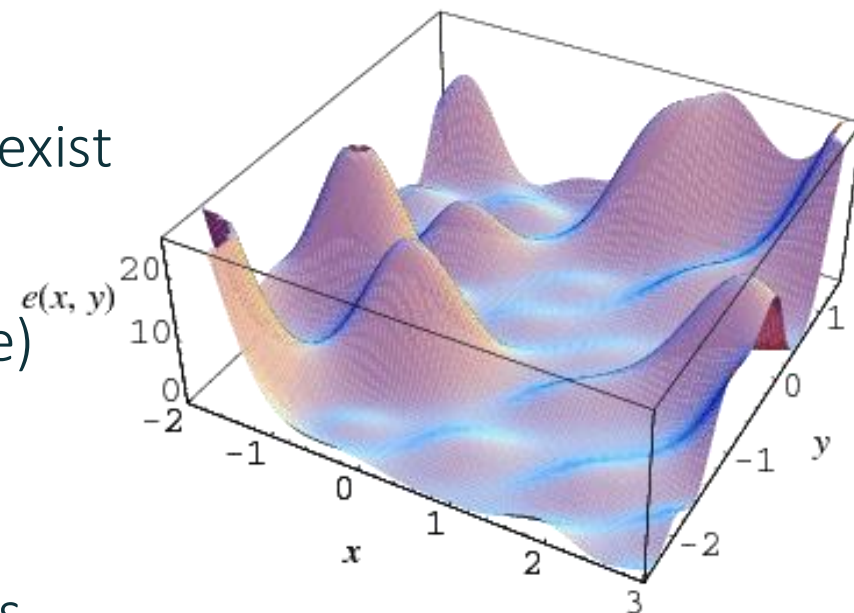
- **Pipes come in discrete diameters**
- **Governing flow equations are non-linear**
- **Systems normally involve thousands of interconnected elements**
 - all mains models
 - small change in one element may have large effect on performance of other elements
 - engineers can try only a small number of designs – small chance of finding “the best”



Single-objective optimisation

Cost (only) optimisation

- **Advantages**
 - single solution - scores better than any other
 - a number of effective optimisation techniques exist
- **Disadvantages**
 - surrogate for genuine trade-off (multi-objective) optimisation of costs, benefits, risk
 - expected value of risk
 - reliability analysis as a surrogate for risk analysis



Multi-objective optimisation

Most real-life problems have multiple, conflicting objectives

Our goal: to generate alternative solutions that ideally provide a range of

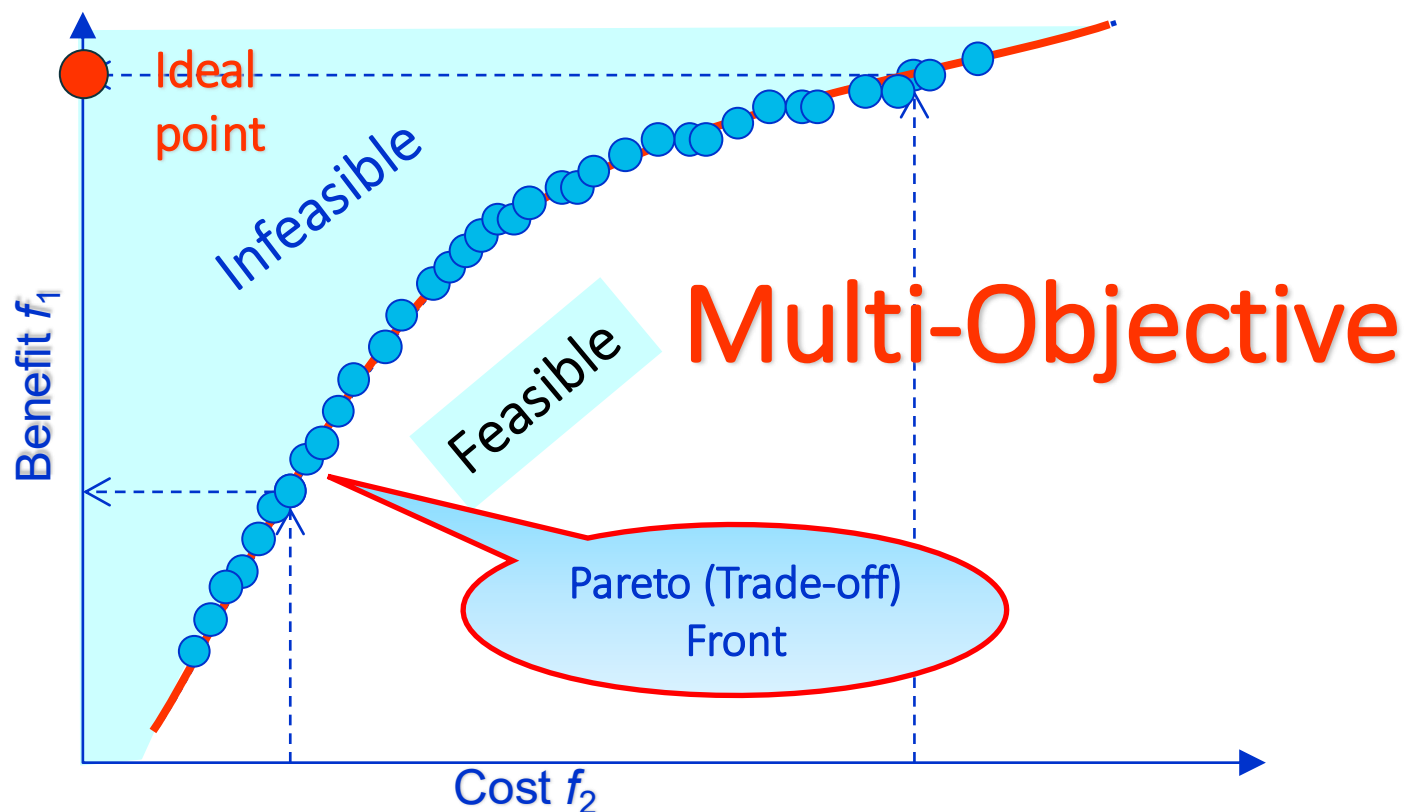
- **costs and**
- **benefits**

Objectives

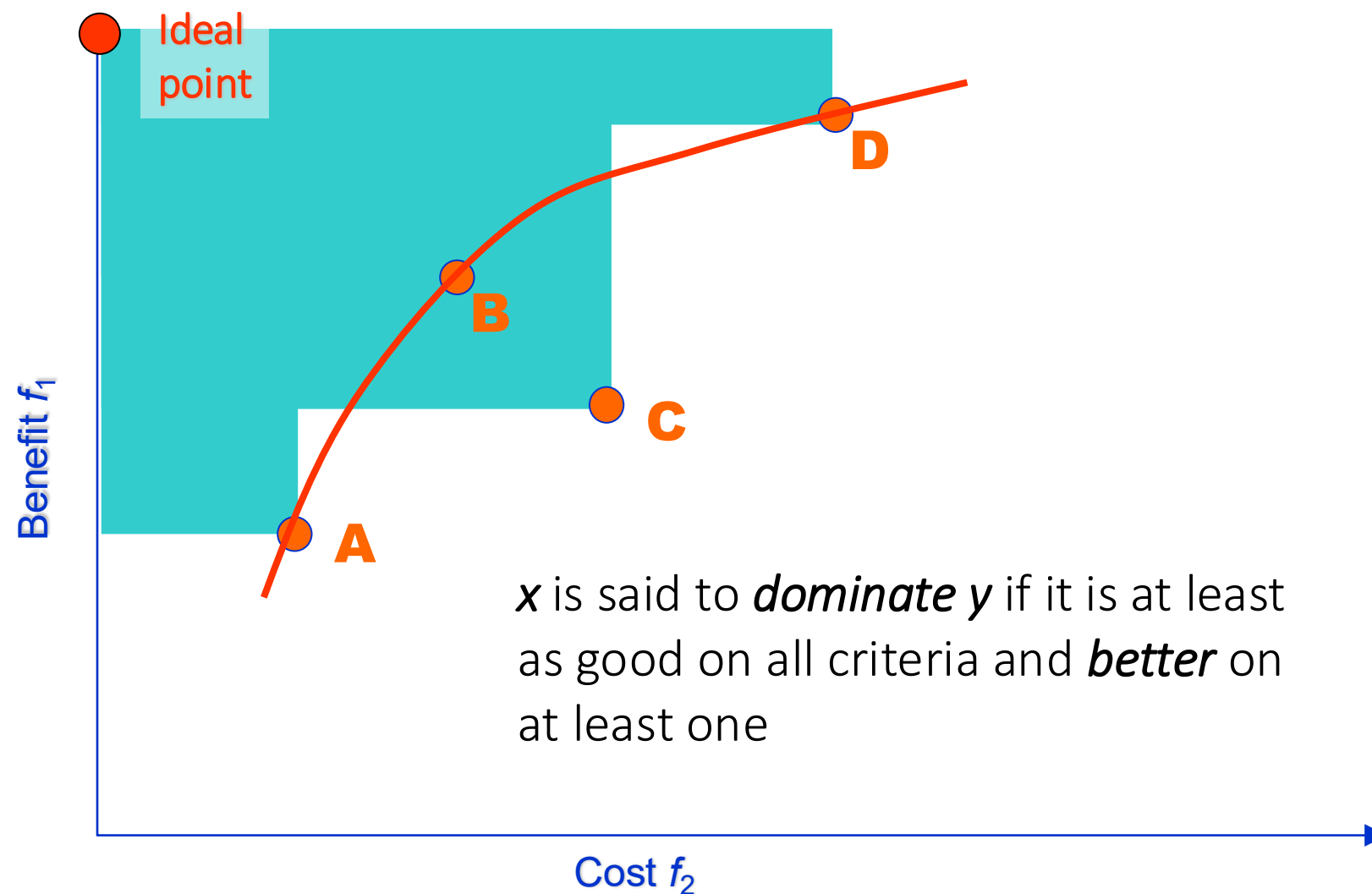
f_1 and f_2

Constraints

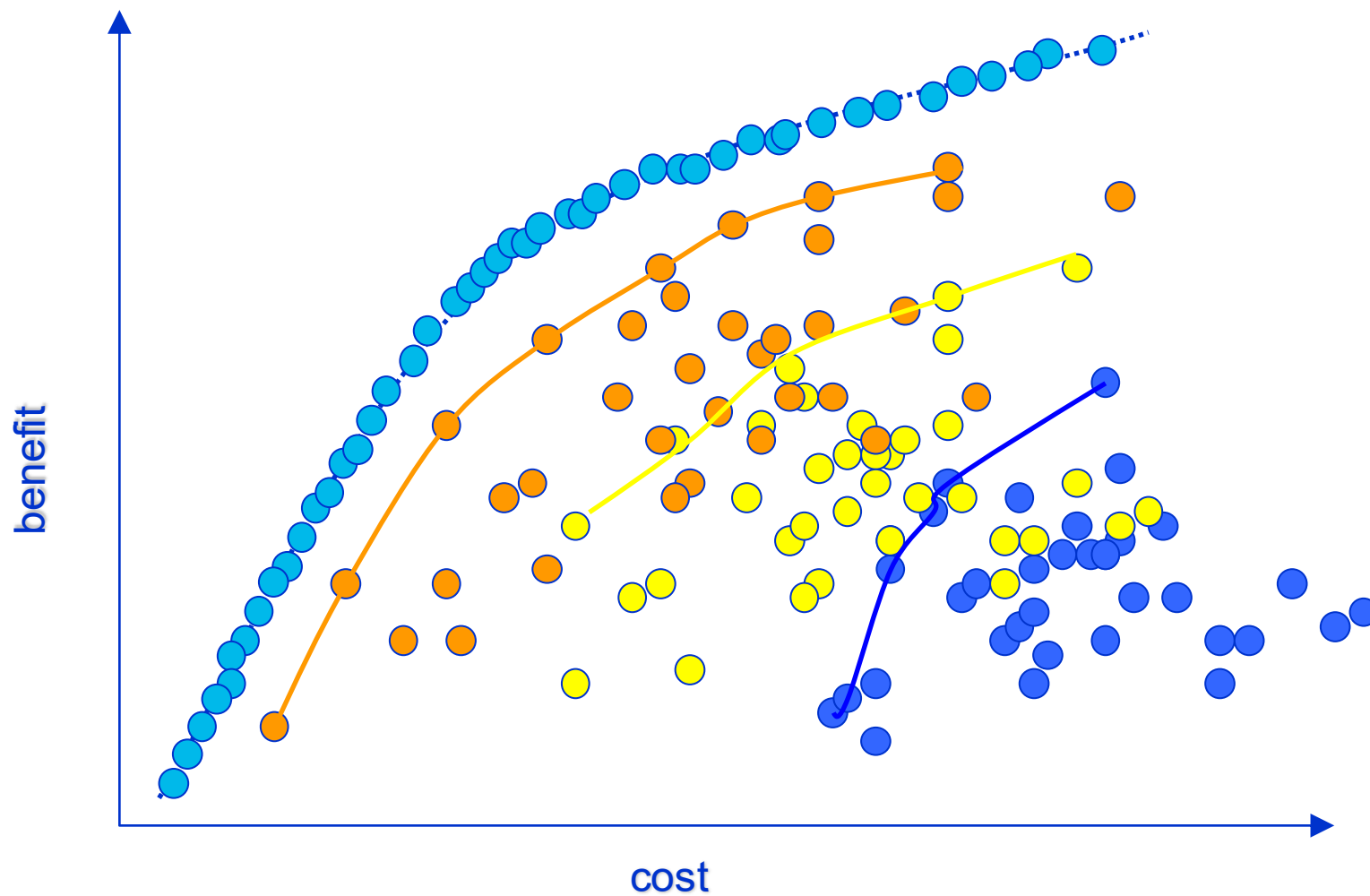
f_1 or f_2



Pareto Dominance



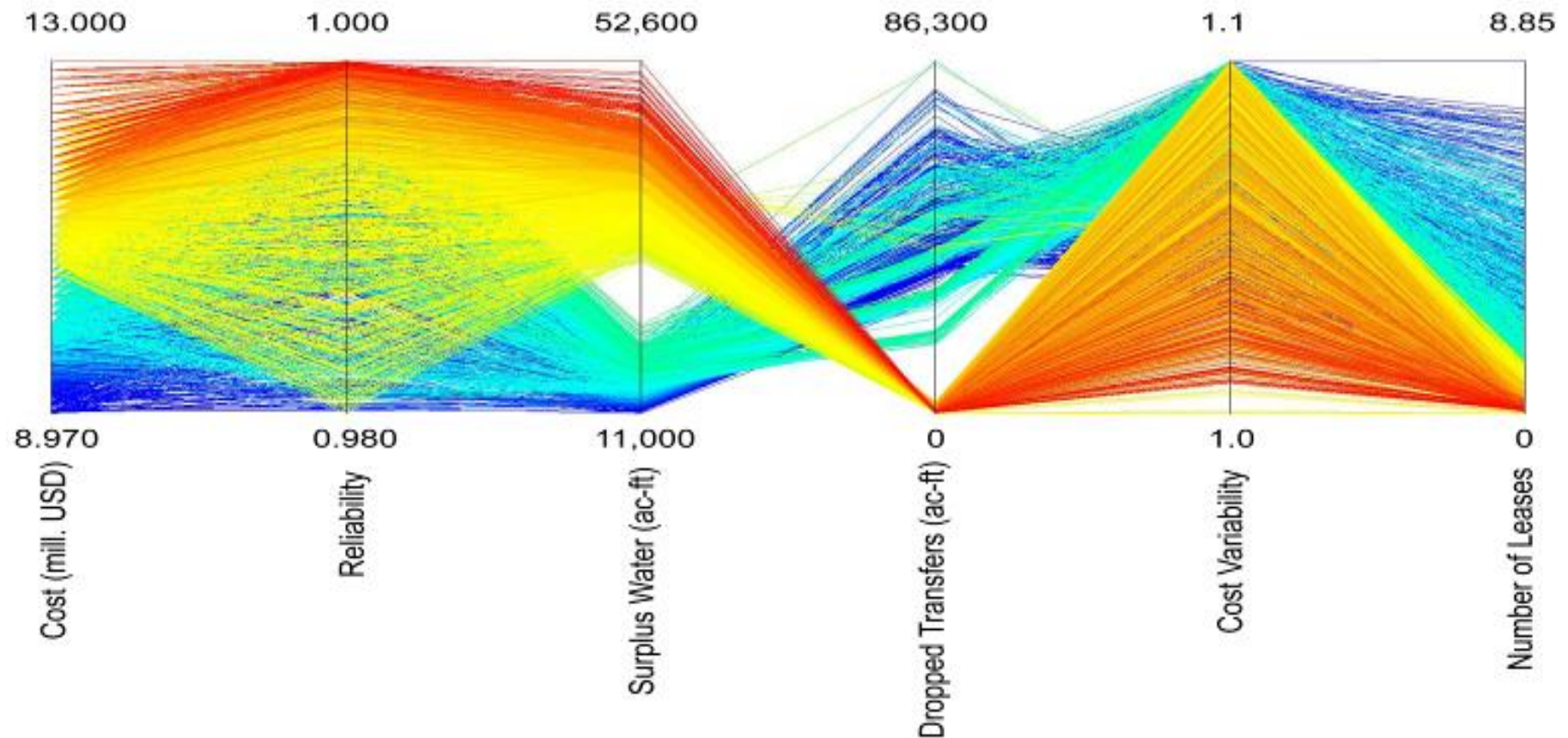
MOGA Pareto set evolution



Multi-objective optimisation

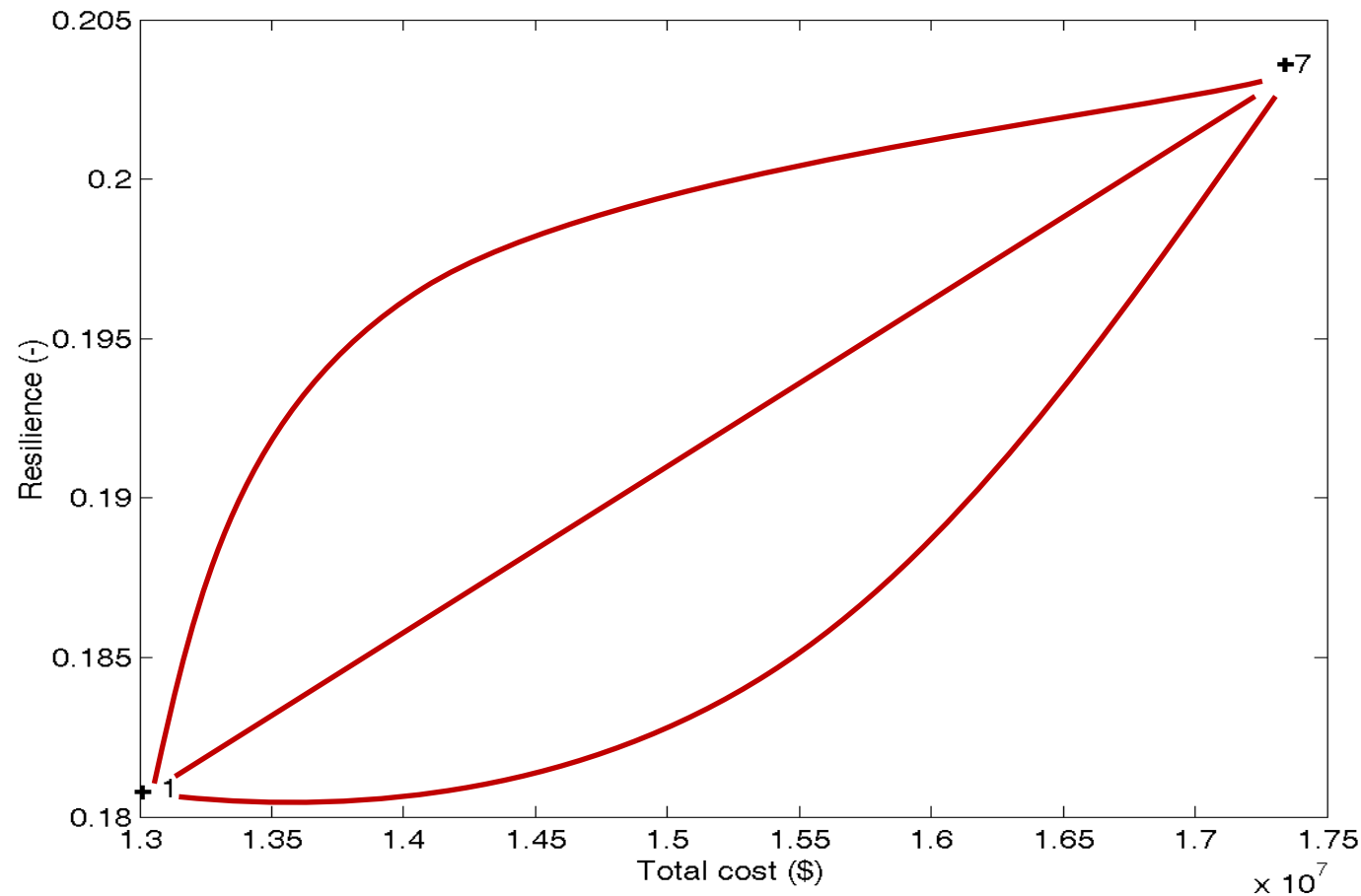
Decision-support tool

Deals directly with a number of performance indicators



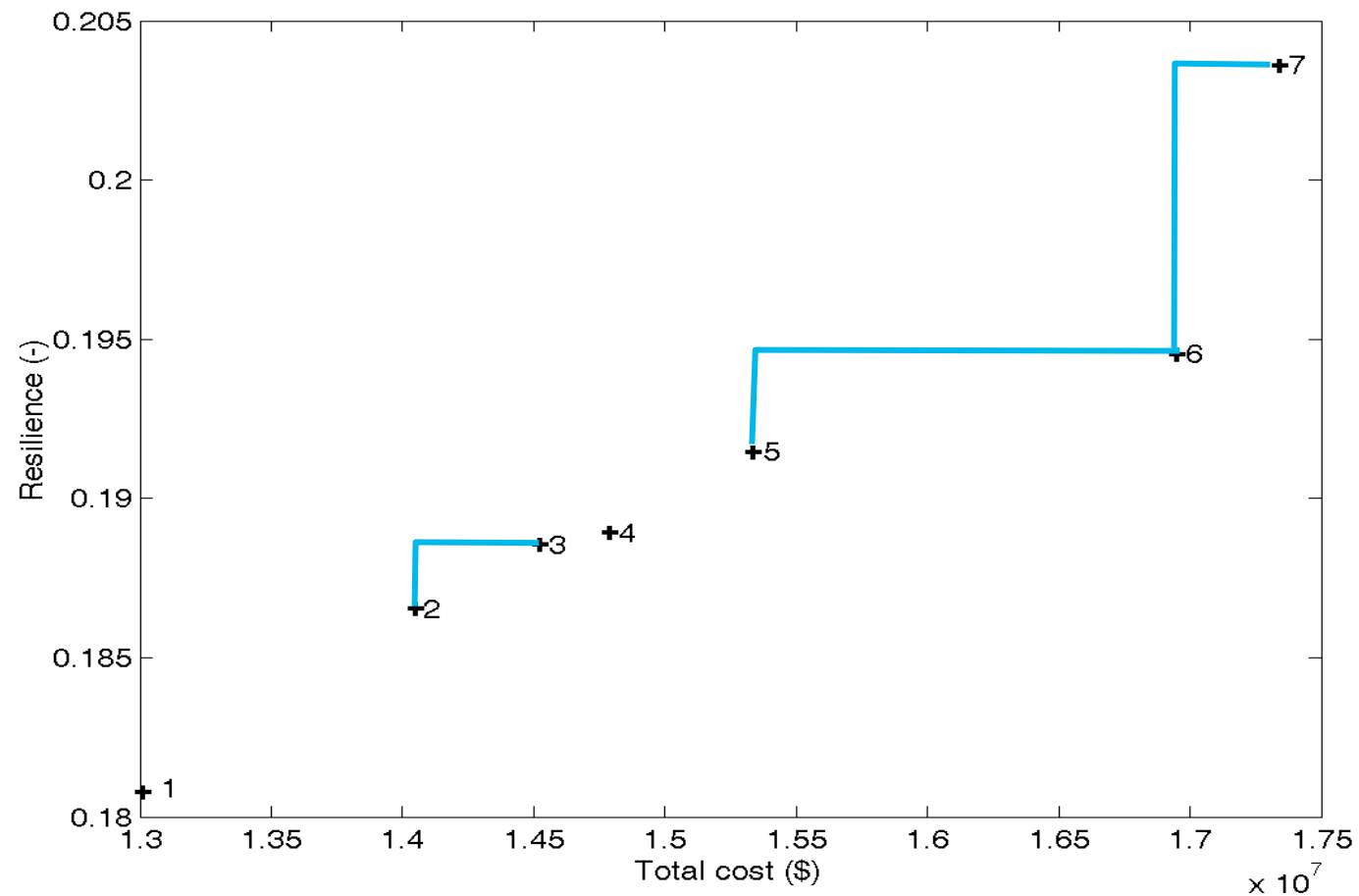
Multi-objective optimisation

Provides the entire Pareto front



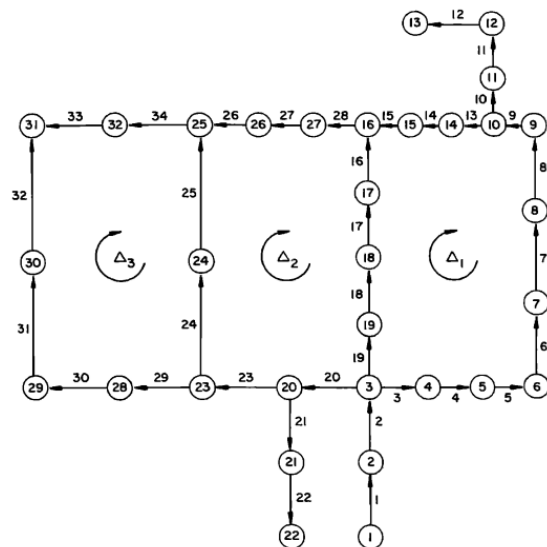
Multi-objective optimisation

Provides entire Pareto front



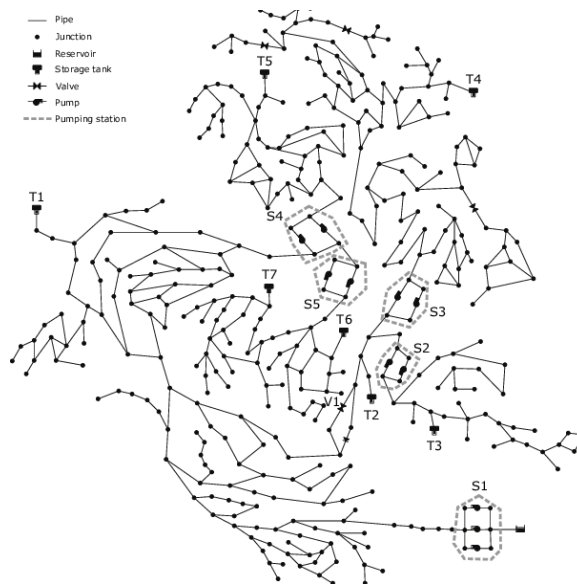
Benchmark Networks

Hanoi



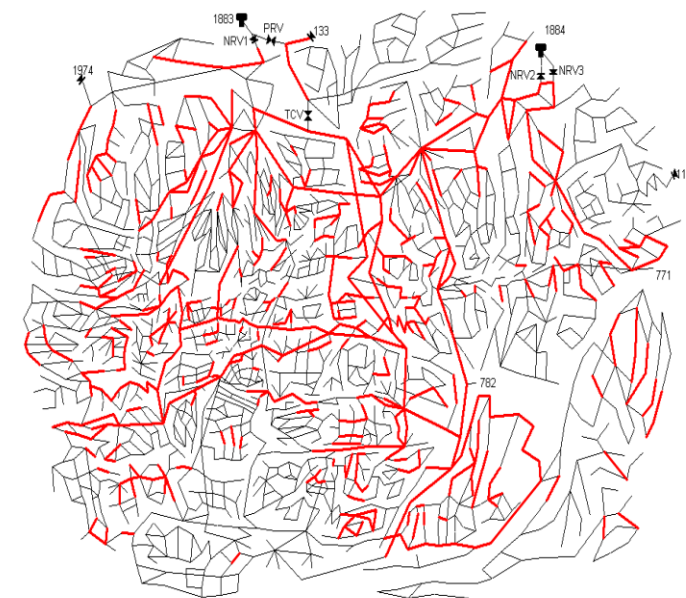
Fujiwara, O. and Khang, D.B., 1990. A two-phase decomposition method for optimal design of looped water distribution networks. *Water Resources Research*, 26(4), pp.539-549.

C-Town



Ostfeld, et al., 2012. Battle of the water calibration networks. *Journal of Water Resources Planning and Management*, 138(5), pp.523-532.

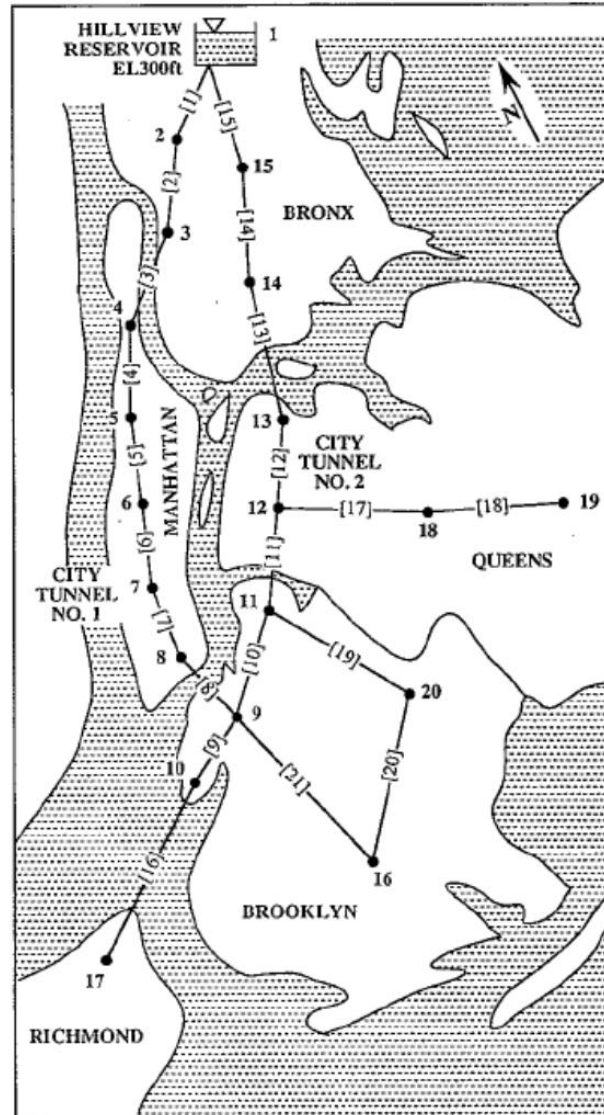
EXNET



Farmani, R., Savic, D.A. and Walters, G.A., 2004. Exnet benchmark problem for multi-objective optimization of large water systems. *Modelling and control for participatory planning and managing water systems*.

New York Tunnels Benchmark (Network Design Tool)

- 1968
- Could not meet future demands
- Minimum pressure requirement



- 21 pipes/tunnels
- Each can be duplicated
- 15 available discrete diameters (plus “do not duplicate”)
- Search space = $16^{21} = 1.9 \times 10^{25}$

Schaake, J. and Lai, D. 1969. “Linear programming and dynamic programming applications to water distribution network design”. In Rep. 116, Dept. of Civ. Engrg., Cambridge, Mass: Massachusetts Inst. of Technol.

Deep Uncertainty

The Precipice

We are planning for a future we cannot reliably predict.



The Capital Burden

€120.8B

in additional capital expenditure required by European service providers by 2030 to maintain compliance.

83.7%

Projected European urbanisation by 2050, straining ageing systems.



The Longevity Trap

"Assets are built to last

50+ years,
yet water receives less than

3% of EU funding."

Decisions made today lock in capital and performance for half a century, making interventions largely irreversible.

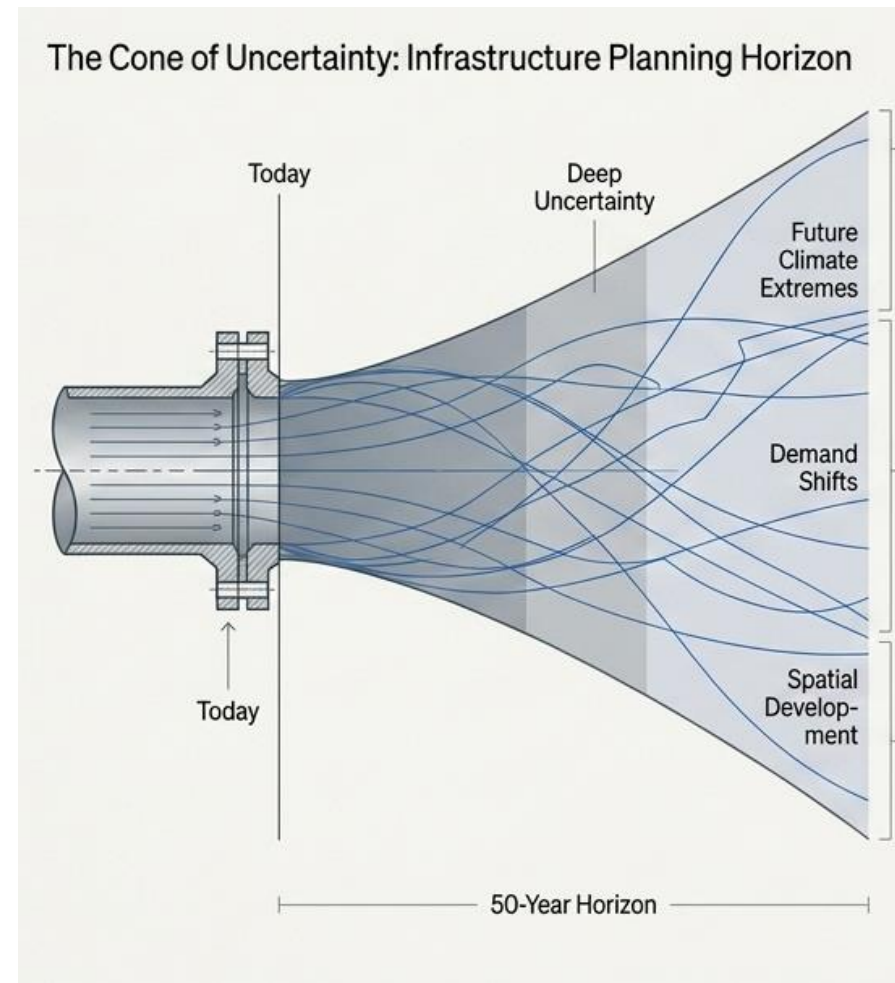


Deep Uncertainty

Future climate extremes, shifting demand, regulations, and spatial development cannot be assigned agreed probabilities. Historical patterns no longer hold.

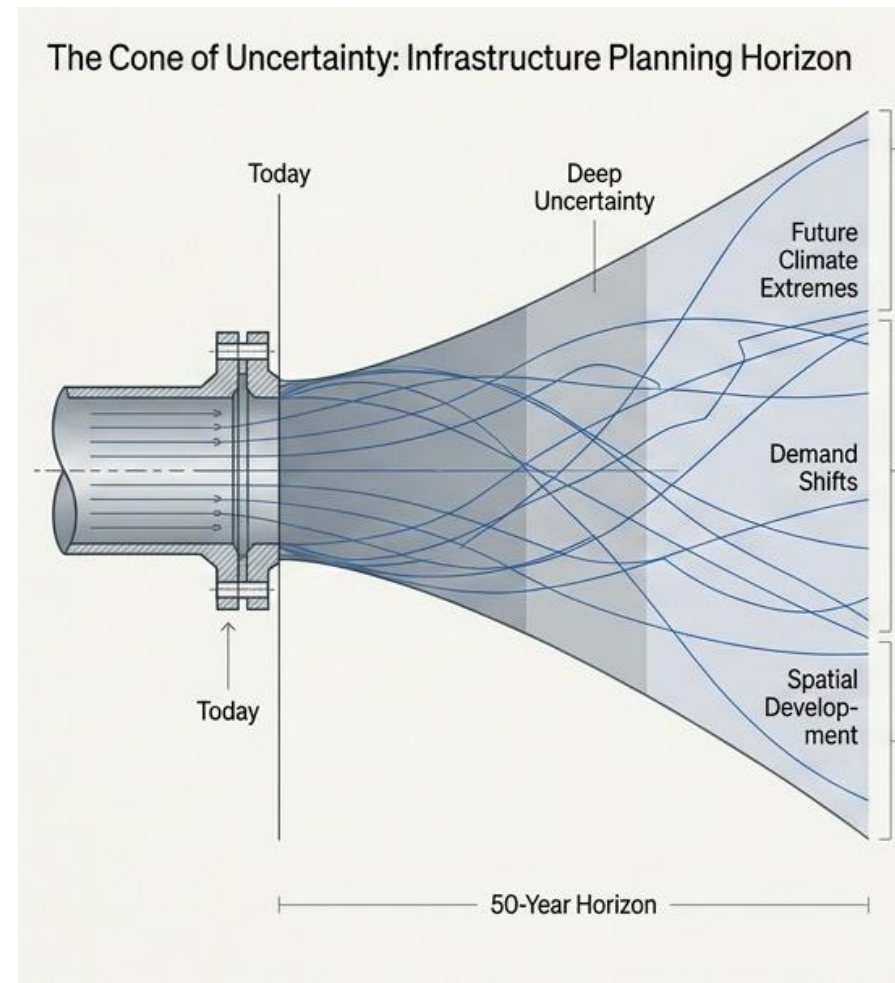
Deep Uncertainty

- Deep uncertainty - we often do not know what the future holds
- Unknown probabilities
- When decision makers assume otherwise, they risk overconfidence, missed opportunities, and pursuing policies brittle to surprise [RAND Corporation]
- A single “best” design may fail under future conditions



Impact of Deep Uncertainty

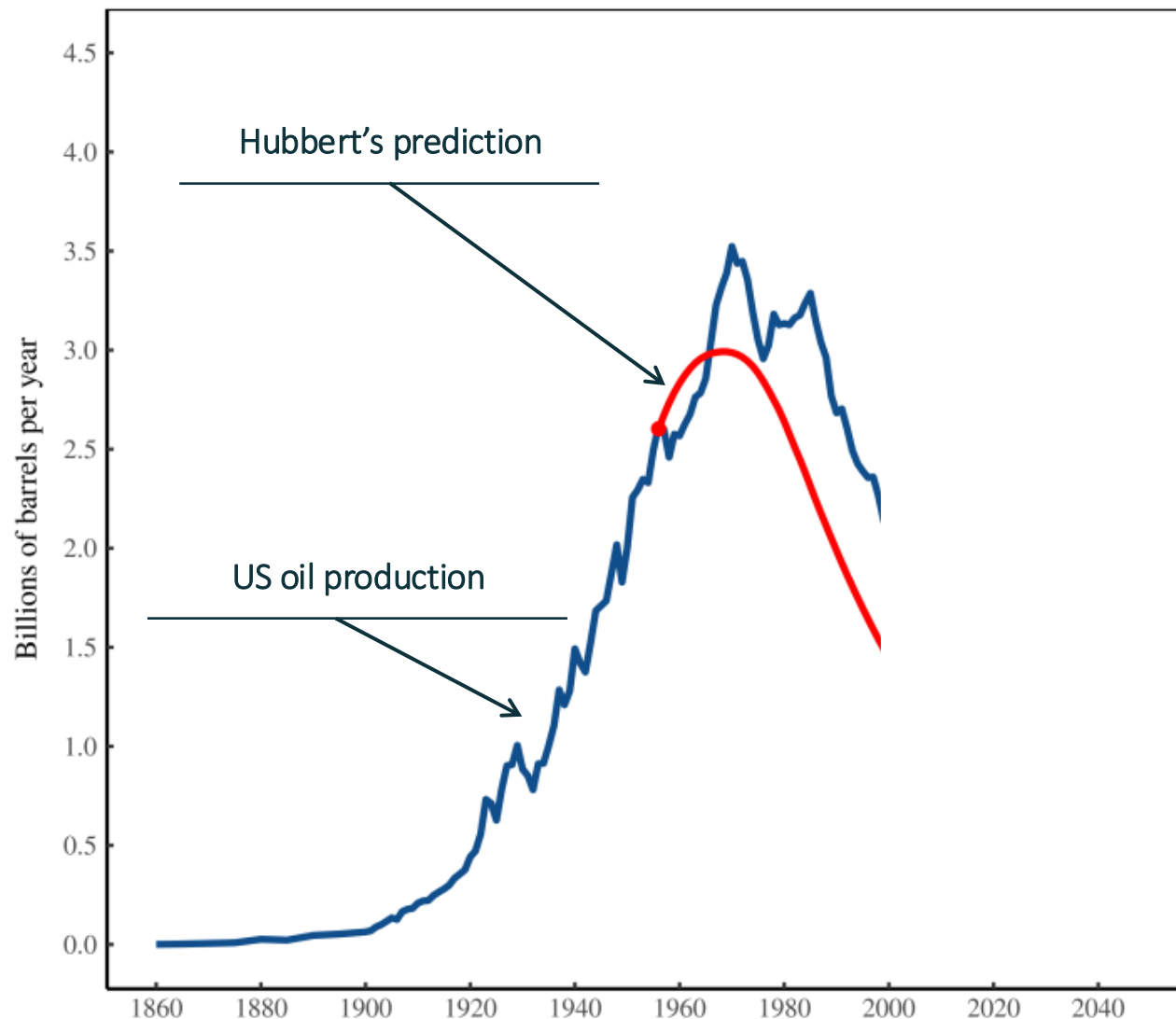
- Future demand is not a single trajectory but a wide range of plausible futures
- The timing of threshold exceedance cannot be predicted probabilistically
- Infrastructure decisions (relocation, redesign, protection) involve irreversible investments with long-term consequences
- Stakeholders must plan for multiple futures rather than one forecast



Peak Oil Prediction

Hubbert predicted in 1956 that

- The US total reserves are 200 billion barrels of oil
- The peak of production would come in 1970

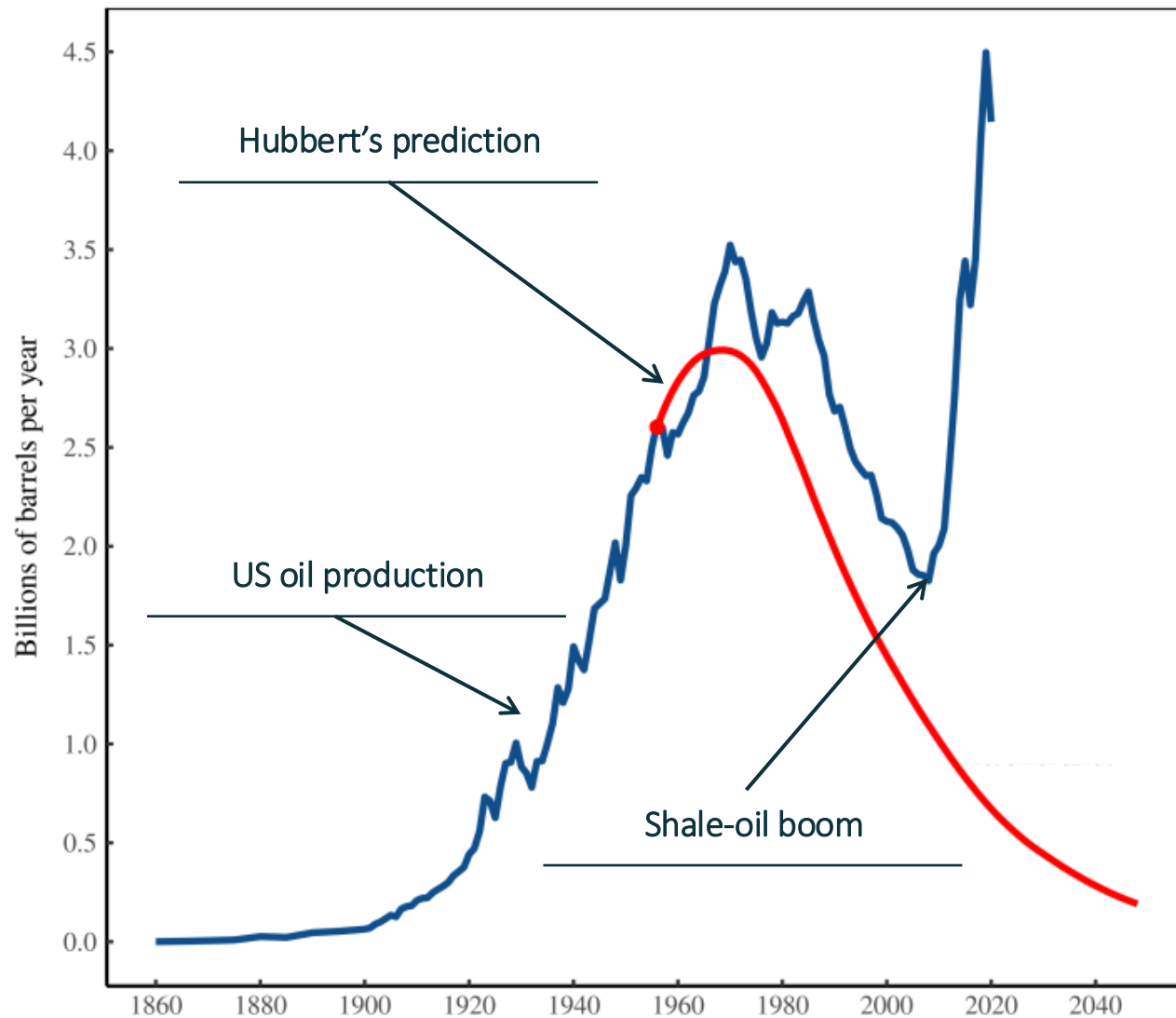


Peak Oil Prediction

Hubbert predicted in 1956 that

- The US total reserves are 200 billion barrels of oil
- The peak of production would come in 1970

In 2008 oil production began to rise





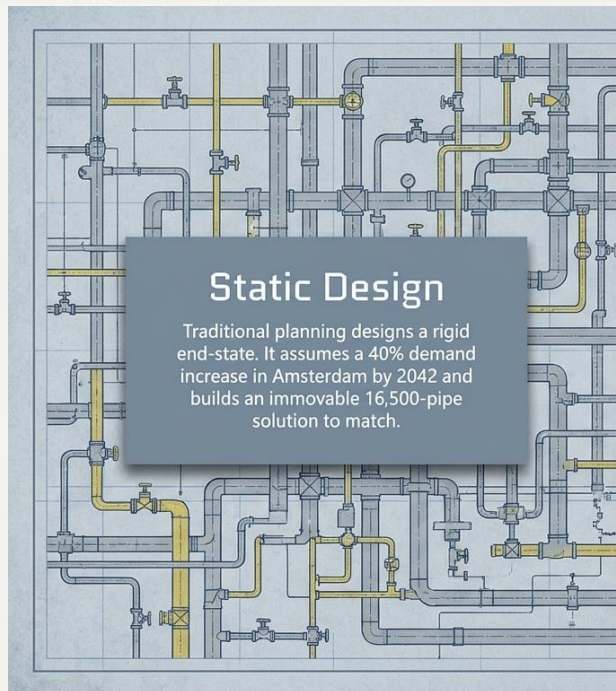
Deep Uncertainty in Water Distribution Design

- **Robust design** works well (i.e., it's resistant) under many different conditions without needing to change.
- **Flexible design** can be adjusted or reconfigured when circumstances change (i.e., does not resist).
- **Adaptive design** learns and evolves over time!

Need robust, flexible, and
adaptive design approaches

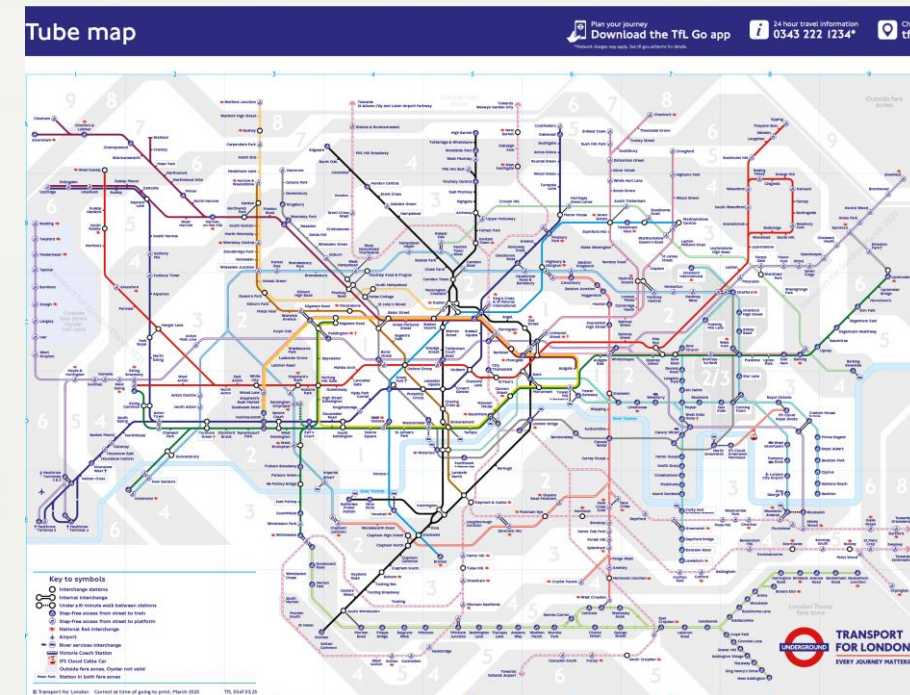
Staged, Robust, and Flexible Design Approaches

From rigid pipes to adaptive journeys.



The Old Model: Water Supply Design

- Fixed, single-path physical architecture.
- Relies entirely on perfectly accurate long-term demand scenarios.
- Fails catastrophically if predictions (e.g., '40% demand increase by 2042') are wrong.



The New Model: Travel Planning

- Multiple alternative routes.
- Complex, multi-line interconnected systems (12 lines, 275 stations).
- Staged, uncertain, and designed for dynamic routing.
- Built around the flexibility to change pathways when a node fails.

Staged, Robust, and Flexible Design Approaches

Staged (Phased) Design

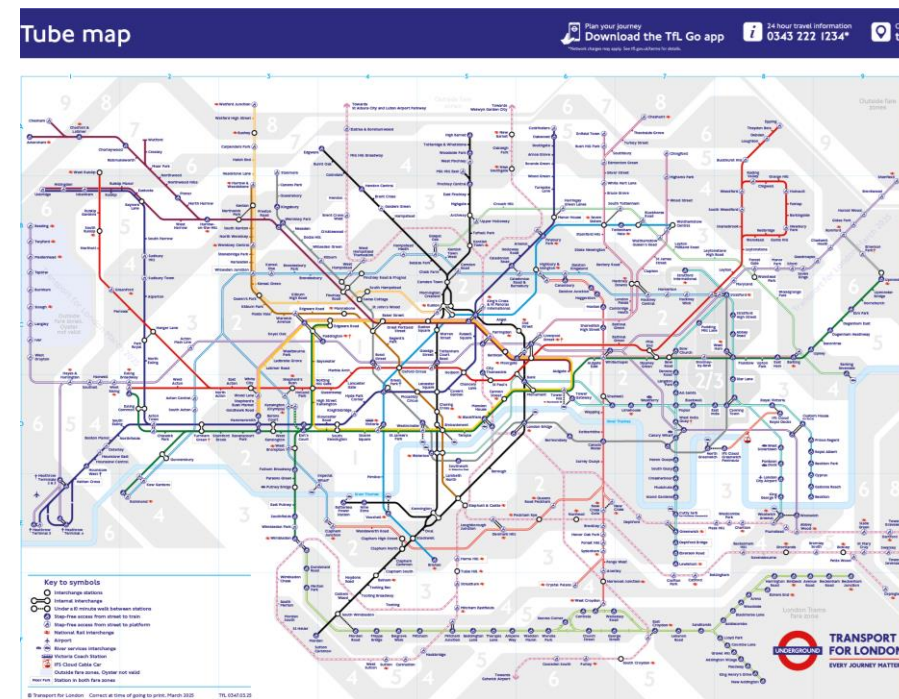
- Build in phases as demand grows
- Reduces upfront cost
- Allows learning and adaptation

Robust Optimisation

- Designs that perform acceptably across many uncertain futures
- Minimise worst-case performance loss
- Trade-off: robustness vs cost

Flexible and Adaptive Design

- Designs that can be modified later at low cost
- Value of flexibility under uncertainty
- Real options thinking (expand, delay, upgrade)



Operational Flexibility

Designing networks that are easier to operate

- Pressure management
- Pump scheduling
- Valve control

Importance of redundancy and alternative flow paths

Flexibility as a design objective

- Ability to respond to failures, demand spikes, and contamination events

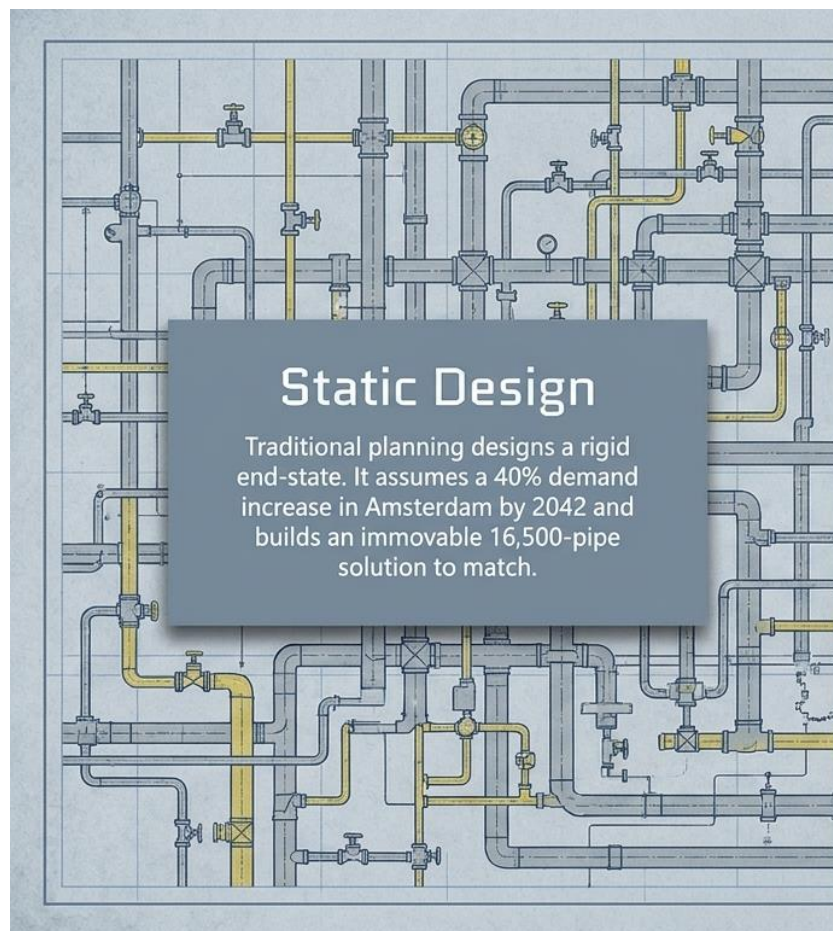
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Strengths and Trade-Offs Across Design Paradigms



Classical (Deterministic/Static) Design

- Strengths: simple, fast, low computational cost
- Weaknesses: fragile under uncertainty

Robust Design

- Strengths: reliable across scenarios
- Weaknesses: may be over-conservative and costly

Flexible/Adaptive Design

- Strengths: future-proof, cost-effective under uncertainty
- Weaknesses: requires more complex modelling and planning

Operationally Flexible Design

- Strengths: improves day-to-day performance and resilience
- Weaknesses: may require more complex infrastructure or control systems

Agentic AI and Reinforcement Learning (RL)

What is agentic AI?

- Systems that take actions, learn from feedback, and adapt

Reinforcement Learning for WDS

- Pump scheduling
- Pressure control
- Leakage management

Strengths

- Learns optimal policies through simulation
- Handles nonlinear, dynamic systems

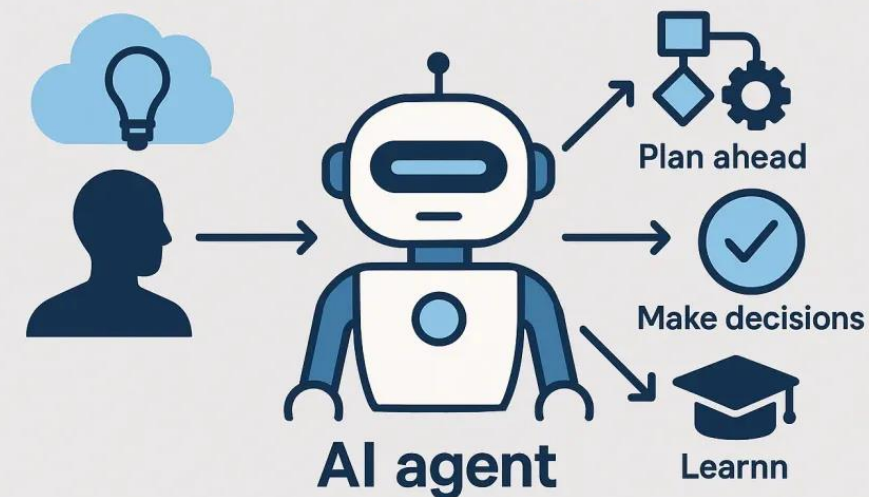
Challenges

- Training time, safety, interpretability
- Need for high-fidelity digital twins

Future potential

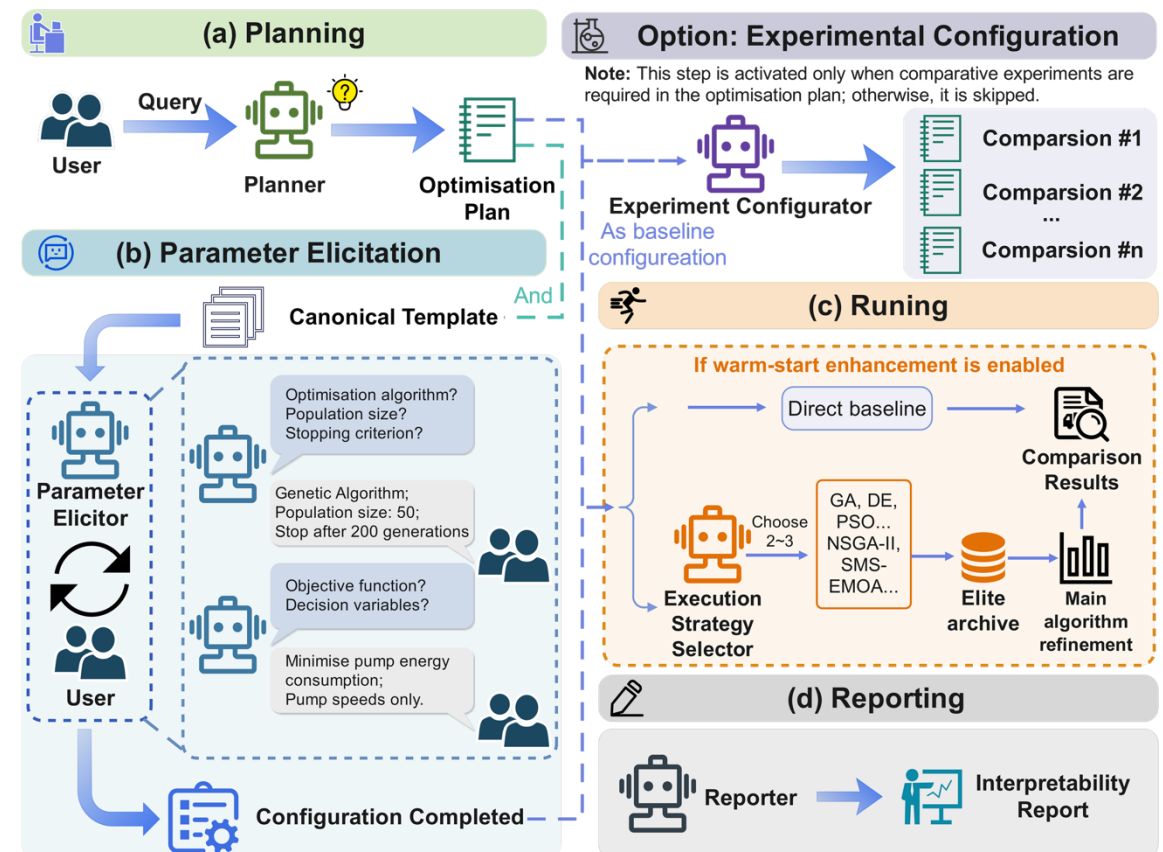
- Autonomous operational optimisation
- AI co-pilots for design and planning

What are AI Agents?



Integrating AI with Classical Optimisation

- Hybrid approaches
- AI for planning, optimization for execution
- LLM for workflow automation
 - Problem formulation
 - Parameter elicitation
 - Scenario generation
 - Reporting and interpretation
- RL for operational optimisation
- Digital twins as the integration platform



Summary & Future Directions

Move from deterministic to uncertainty-aware design

- Rigid, fragile, prone to break

Combine robust, flexible and operational perspectives

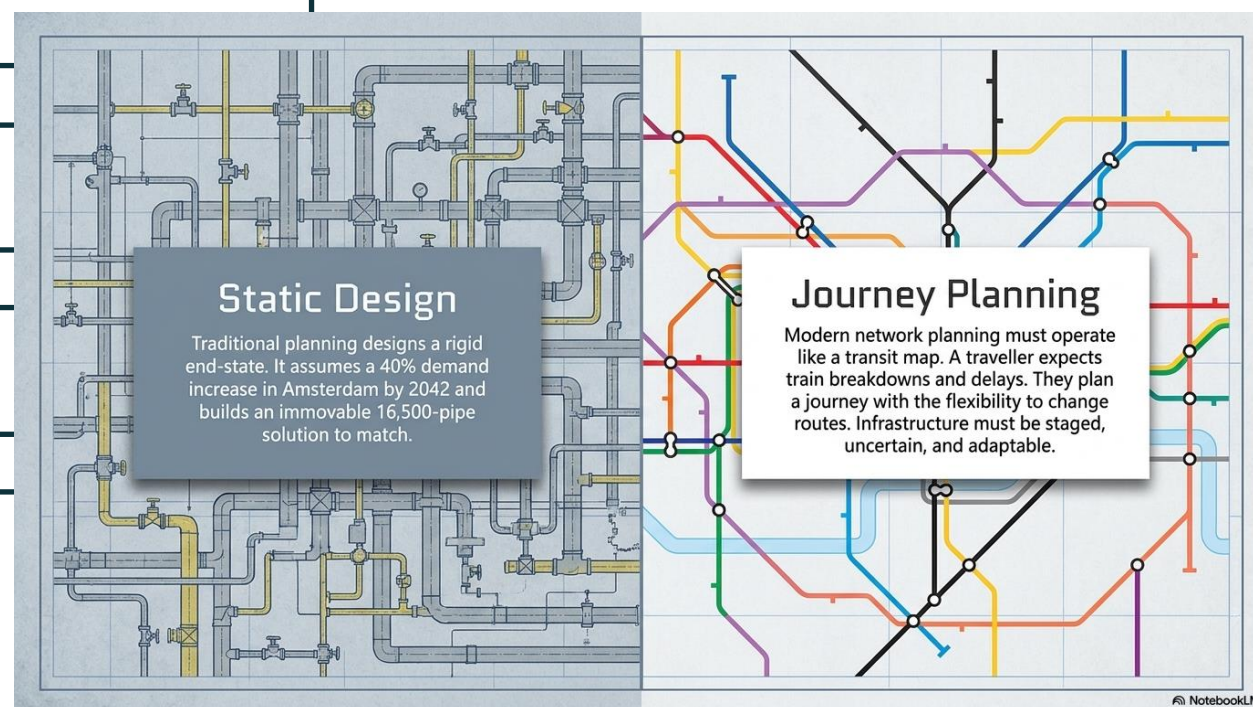
- Long-term sustainability

AI, agentic AI and RL offer new capabilities

- Human-in-the-loop = oversight

Future

- AI-assisted design
- Real-time optimisation
- Fully integrated digital twins
- Human-in-the-loop decision system





Future?

A system-wide “AI Co-Pilot” for water utilities that is continuously learning and working in sync with the human



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Thank you!